NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

Unit	Unit Title	≈ # of Weeks	Date Range
1	***Incorporate WIM - Week of Inspirational Math*** Equations & Introduction to Functions	4 Weeks (20)	8/13 - 9/10
2	Linear Functions	9 Weeks (44)	9/11 - 11/14
	DPI NC Check-In 1	***Suggested after Unit 2***	
3	Introduction to Exponential Functions	6 Weeks (32)	11/17 - 1/16
4	Introduction to Quadratic Functions and Equations	6 Weeks (34)	1/20 - 3/6
5	Systems of Equations and Inequalities	5 Weeks (18)	3/16 - 4/10
	DPI NC Check-In 2	***Suggested after Unit 5***	
6	Descriptive Statistics	2 Weeks (10)	4/13 - 4/24
	Review **EOC Exam**	**(10)** **(10)**	**Review: 4/27 – 5/8** **EOC: 5/11 – 5/22**

The number of suggested weeks for each unit includes assessment and review days.
Each unit will need flexibility based on the need for reteaching and re-engagement.

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

Standards for Mathematical Practice	
1. Make sense of problems and persevere in solving them.	5. Use appropriate tools strategically.
2. Reason abstractly and quantitatively.	6. Attend to precision.
3. Construct viable arguments and critique reasoning of others.	7. Look for and make use of structure.
4. Model with mathematics.	8. Look for and express regularity in repeated reasoning.

Unit 1: Introduction to Functions & Equations	**≈ # of Weeks: 4**

Concepts to Integrate: Apply all to real world applications.

- Create linear equations and inequalities that describe numbers or relationships.
- Understand solving equations and inequalities in one variable as a process of reasoning and explain the reasoning.
- Solve equations and inequalities in one variable.
- Understand the concept of a function and use function notation with linear ***and*** non-linear functions.
- Interpret functions that arise in applications in terms of the context.

SMPs: *See Unit Planning Organizer*

Construct expressions, equations, and inequalities from a given context and determine the appropriateness of the solution.	
NC.M1.A-SSE.1a	Interpret expressions that represent a quantity in terms of its context. Identify and interpret parts of a linear, exponential, or quadratic expression, including terms, factors, coefficients, and exponents.
NC.M1.A-REI.3	Solve linear equations and inequalities in one variable.

NC.M1.A-REI.1	Understand solving equations as a process of reasoning and explain the reasoning. Justify a chosen solution method and each step of the solving process for linear and quadratic equations using mathematical reasoning.

NC.M1.A-REI.12	Represent the solutions of a **linear inequality** or a system of linear inequalities graphically as a region of the plane.
NC.M1.A-CED.1	Create equations that describe numbers or relationships. Create equations and inequalities in one variable that represent linear, exponential, and quadratic relationships and use them to solve problems.
NC.M1.A-CED.4	Create equations that describe numbers or relationships. Solve for a quantity of interest in formulas used in science and mathematics using the same reasoning as in solving equations.

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

Distinguish key features of a function given multiple representations.	
NC.M1.F-IF.1	Understand the concept of a function and use function notation. Build an understanding that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range by recognizing that: ● if f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x ● the graph of f is the graph of the equation $y = f(x)$.
NC.M1.F-IF.2	Use function notation to evaluate linear, quadratic, and exponential functions for inputs in their domains, and interpret statements that use function notation in terms of a context.
NC.M1.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities, including: intercepts; intervals where the function is increasing, decreasing, positive, or negative; and maximums and minimums.
NC.M1.F-IF.6	Interpret functions that arise in applications in terms of the context. Calculate and interpret the average rate of change over a specified interval for a function presented numerically, graphically, and/or symbolically.

8th Grade Standards:

Analyze and solve linear equations and pairs of simultaneous linear equations.	
NC.8.EE.7	Solve real-world and mathematical problems by writing and solving equations and inequalities in one variable. ● Recognize linear equations in one variable as having one solution, infinitely many solutions, or no solutions. ● Solve linear equations and inequalities including multi-step equations and inequalities with the same variable on both sides.
Define, evaluate, and compare functions.	
NC.8.F.1	Understand that a function is a rule that assigns to each input exactly one output. ● Recognize functions when graphed as the set of ordered pairs consisting of an input and exactly one corresponding output. ● Recognize functions given a table of values or a set of ordered pairs.
Use functions to model relationships between quantities.	
NC.8.F.5	Qualitatively analyze the functional relationship between two quantities. ● Analyze a graph determining where the function is increasing or decreasing; linear or non-linear. ● Sketch a graph that exhibits the qualitative features of a real-world function.

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

Unit 2: Linear Functions	**≈ # of Weeks: 10**
NC DPI NC Check-In #1	

Concepts to Integrate: Apply all to real world applications.

- Interpret the structure of linear expressions.
- Create linear equations that describe numbers or relationships.
- Represent and solve linear equations and inequalities graphically
- Understand the concept of a linear function and use function notation.
- Interpret linear functions that arise in applications in terms of the context.
- Analyze linear functions using different representations.
- Build a linear function that models a relationship between two quantities.
- Interpret expressions for linear functions in terms of the situation they model.
- Recognize an arithmetic sequence as a linear function and write a recursive and explicit rule.
- Construct and compare linear models and solve problems.
- Summarize and compare linear models and solve problems.
- Assess the line of best fit.
- Use geometric properties to classify and prove figures.
- Use coordinates to prove simple geometric theorems with polygons algebraically.
- Use slope criteria of parallel and perpendicular lines to solve problems.
- Find the endpoint and midpoint of given coordinates of a line segment.

SMPs: *See Unit Planning Organizer*

Identify, create, and graph linear equations and inequalities and interpret their key features.	
NC.M1.A-SSE.1 **NC.M1.A-SSE.1a**	Interpret expressions that represent a quantity in terms of its context. a. Identify and interpret parts of a **linear**, exponential, or quadratic expression, including terms, factors, coefficients, and exponents.
NC.M1.A-SSE.1b	b. Interpret a **linear**, exponential, or quadratic expression made of multiple parts as a combination of entities to give meaning to an expression.
NC.M1.A-CED.1	Create equations and inequalities in one variable that represent linear, exponential, and quadratic relationships and use them to solve problems.
NC.M1.A-REI.12	Represent the solutions of a linear inequality or a system of linear inequalities graphically as a region of the plane.

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

NC.M1.A-CED.2	Create and graph equations in two variables to represent **linear**, exponential, and quadratic relationships between quantities.
NC.M1.F-BF.1a	Build a function that models a relationship between two quantities. Write a function that describes a relationship between two quantities. Build **linear** and exponential functions, including arithmetic and geometric sequences, given a graph, a description of a relationship, or two ordered pairs (include reading these from a table).

NC.M1.A-REI.10	Understand that the graph of a two variable equation represents the set of all solutions to the equation.
NC.M1.A-REI.11	Build and understanding of why the x-coordinates of the points where the graphs of **two linear**, exponential, or quadratic equations $y = f(x)$ *and* $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$ and approximate solutions using a graphing technology or successive approximations with a table of values.
NC.M1.G-GPE.5	Use coordinates to prove simple geometric theorems algebraically. Use coordinates to prove the slope criteria for parallel and perpendicular lines and use them to solve problems. Determine if two lines are parallel, perpendicular, or neither. Find the equation of a line parallel or perpendicular to a given line that passes through a given point.
Determine the explicit and recursive formula for given arithmetic sequence.	
NC.M1.F-IF.3	Recognize that recursively and explicitly defined sequences are functions whose domain is a subset of the integers, **the terms of an arithmetic sequence are a subset of the range of a linear function,** and the terms of a geometric sequence are a subset of the range of an exponential function.
NC.M1.F-BF.2	Build a function that models a relationship between two quantities. Translate between explicit and recursive forms of **arithmetic** and geometric sequences and use both to model situations.
NC.M1.A-REI.1	Understand solving equations as a process of reasoning and explain the reasoning. Justify a chosen solution method and each step of the solving process for linear and quadratic equations using mathematical reasoning.
Understand and compare key features of linear functions.	
NC.M1.F-LE.5	Interpret expressions for functions in terms of the situation they model. Interpret the parameters a and b in a linear function f(x)= ax+b or an exponential function $g(x) = ab^x$ in terms of a context.
NC.M1.F-IF.5	Interpret a function in terms of the context by relating its domain and range to its graph and, where applicable, to the quantitative relationship it describes.
NC.M1.S-ID.9	Interpret linear models. Distinguish between association and causation.
NC.M1.F-IF.6	Calculate and interpret the **average rate of change** over a specified interval for a function presented numerically, graphically, and/or symbolically.
NC.M1.F-IF.7	Analyze **linear**, exponential, and quadratic functions by generating different representations, by hand in simple cases and using technology for more complicated cases, to show key features, including: domain and range; rate of change; intercepts; intervals where the function is increasing, decreasing, positive, or negative; maximums and minimums; and end behavior.
NC.M1.F-IF.9	Compare key features of two functions (**linear**, quadratic, or exponential) each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
Assess the line of best fit for a given set of data by using the correlation coefficient, residuals, and the least squares regression line.	

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

NC.M1.S-ID.6a	Summarize, represent, and interpret data on two categorical and quantitative variables. Represent data on two quantitative variables on a scatter plot, and describe how the variables are related. Fit a least squares regression line to linear data using technology. Use the fitted function to solve problems.
NC.M1.S-ID.6b	Summarize, represent, and interpret data on two categorical and quantitative variables. Represent data on two quantitative variables on a scatter plot, and describe how the variables are related. Assess the fit of a linear function by analyzing residuals.
NC.M1.S-ID.7	Interpret linear models. Interpret in context the rate of change and the intercept of a linear model. Use the linear model to interpolate and extrapolate predicted values. Assess the validity of a predicted value.
NC.M1.S-ID.8	Interpret linear models. Analyze patterns and describe relationships between two variables in context. Using technology, determine the correlation coefficient of bivariate data and interpret it as a measure of the strength and direction of a linear relationship. Use a scatter plot, correlation coefficient, and a residual plot to determine the appropriateness of using a linear function to model a relationship between two variables.
Use geometric properties to classify & prove figures in the coordinate plane.	
NC.M1.G-GPE.5	Use coordinates to prove simple geometric theorems algebraically. Use coordinates to prove the slope criteria for parallel and perpendicular lines and use them to solve problems. Determine if two lines are parallel, perpendicular, or neither. Find the equation of a line parallel or perpendicular to a given line that passes through a given point.
NC.M1.G-GPE.6	Use coordinates to prove simple geometric theorems algebraically. Use coordinates to find the midpoint or endpoint of a line segment.
NC.M1.G-GPE.4	Use coordinates to prove simple geometric theorems algebraically. Use coordinates to solve geometric problems involving polygons algebraically. Use coordinates to compute perimeters of polygons and areas of triangles and rectangles. Use coordinates to verify algebraically that a given set of points produces a particular type of triangle or quadrilateral.

8th Grade Standards:

Define, evaluate, and compare functions.	
NC.8.F.2	Compare properties of two linear functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).
NC.8.F.3	Identify linear functions from tables, equations, and graphs.
Use functions to model relationships between quantities.	

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

NC.8.F.4	Analyze functions that model linear relationships. • Understand that a linear relationship can be generalized by $y = mx + b$. • Write an equation in slope-intercept form to model a linear relationship by determining the rate of change and the initial value, given at least two *(x, y)* values or a graph. • Construct a graph of a linear relationship given an equation in slope-intercept form. • Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of the slope and y-intercept of its graph or a table of values.

Additional 8th Grade Standards:

Investigate patterns of association in bivariate data.	
NC.8.SP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Investigate and describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.
NC.8.SP.2	Model the relationship between bivariate quantitative data to: • Informally fit a straight line for a scatter plot that suggests a linear association. • Informally assess the model fit by judging the closeness of the data points to the line.
NC.8.SP.3	Use the equation of a linear model to solve problems in the context of bivariate quantitative data, interpreting the slope and y- intercept.
Understand and apply the Pythagorean Theorem.	
NC.8.G.6	Explain the Pythagorean Theorem and its converse.
NC.8.G.7	Apply the Pythagorean Theorem and its converse to solve real-world and mathematical problems.
NC.8.G.8	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.

Unit 3: Introduction to Exponential Functions	**≈ of # Weeks: 6**
Concepts to Integrate: Apply all to real world applications. • Extend the properties of exponents to integer exponents. • Create exponential equations that describe numbers or relationships. • Represent and solve a system linear and exponential equations and graphically • Understand the concept of an exponential function and use function notation.	

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

- Interpret exponential functions that arise in applications in terms of the context.
- Analyze exponential functions using different representations.
- Build an exponential function that models a relationship between two quantities.
- Construct and compare linear and exponential models and solve problems.
- Interpret expressions for exponential functions in terms of the situation they model.
- Summarize, represent and interpret data on two categorical and quantitative variables.
- Recognize a geometric sequence as an exponential function and write a recursive and explicit rule.

SMPs: *See Unit Planning Organizer*

Understand and apply the properties of exponents.	
NC.M1.N-RN.2	Extend the properties of exponents. Rewrite algebraic expressions with integer exponents using the properties of exponents.
Determine the explicit and recursive formula for given geometric sequences.	
NC.M1.F-IF.3	Recognize that recursively and explicitly defined sequences are functions whose domain is a subset of the integers, the terms of an arithmetic sequence are a subset of the range of a linear function, and **the terms of a geometric sequence are a subset of the range of an exponential function.**
NC.M1.F-BF.2	Translate between explicit and recursive forms of arithmetic and **geometric sequences** and use both to model situations.
Evaluate, create, and interpret exponential functions in context.	
NC.M1.F-IF.2	Understand the concepts of a functions and use function notation. Use function notation to evaluate linear, quadratic, and exponential functions for inputs in their domains, and interpret statements that use function notation in terms of a context.
NC.M1.F-IF.4	Interpret functions that arise in applications in terms of the context. Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities, including: intercepts; intervals where the function is increasing, decreasing, positive, or negative; and maximums and minimums.
NC.M1.A-CED.1	Create equations that describe numbers or relationships. Create equations and inequalities in one variable that represent linear, **exponential**, and quadratic relationships and use them to solve problems.
NC.M1.A-CED.2	Create equations that describe numbers or relationships. Create and graph equations in two variables to represent linear, **exponential**, and quadratic relationships between quantities.
NC.M1.A-REI.10	Represent and solve equations and inequalities graphically. Understand that the graph of a two variable equation represents the set of all solutions to the equation.
NC.M1.F-LE.5	Interpret expressions for functions in terms of the situation they model. Interpret the parameters a and b in a linear function $f(x) = ax + b$ or an exponential function $g(x) = ab^x$ in terms of a context.
NC.M1.F-IF.6	Interpret functions that arise in applications in terms of the context. Calculate and interpret the average rate of change over a specified interval for a function presented numerically, graphically, and/or symbolically.
Identify situations and practical domains for exponential functions.	

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

NC.M1.F-LE.1	Identify situations that can be modeled with linear and exponential functions, and justify the most appropriate model for a situation based on the rate of change over equal intervals.
NC.M1.F-IF.5	Interpret a function in terms of the context by relating its domain and range to its graph and, where applicable, to the quantitative relationship it describes.
NC.M1.F-IF.7	Analyze linear, **exponential**, and quadratic functions by generating different representations, by hand in simple cases and using technology for more complicated cases, to show key features, including: domain and range; rate of change; intercepts; intervals where the function is increasing, decreasing, positive, or negative; maximums and minimums; and end behavior.
Compare, interpret, and explain key features of exponential functions.	
NC.M1.F-IF.8b	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities, including: intercepts; intervals where the function is increasing, decreasing, positive, or negative; and maximums and minimums.
NC.M1.F-LE.3	Construct and compare **linear and exponential** models and solve problems. Compare the end behavior of **linear, exponential**, and quadratic functions using graphs and tables to show that a quantity increasing **exponentially** eventually exceeds a quantity increasing **linearly** or quadratically.
NC.M1.A-SSE.1a	Interpret the structure of expressions. Identify and interpret parts of a linear, exponential, or quadratic expression, including terms, factors, coefficients, and exponents.
NC.M1.F-IF.9	Compare key features of two functions (linear, quadratic, or exponential) each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
NC.M1.A-REI.11	Build an understanding of why the *x*-coordinates of the points where the graphs of two linear, **exponential**, or quadratic equations intersect are the solutions of the equation and approximate solutions using a graphing technology or successive approximations with a table of values.
Write and apply exponential functions given multiple representations.	
NC.M1.F-BF.1a	Build linear and **exponential** functions, including arithmetic and **geometric** sequences, given a graph, a description of a relationship, or two ordered pairs (include reading these from a table).
NC.M1.F-BF.1b	Build a function that models a relationship between two quantities by combining linear, **exponential**, or quadratic functions with addition and subtraction or two linear functions with multiplication.
NC.M1.S.ID.6c	Represent data on two quantitative variables on a scatter plot, and describe how the variables are related. Fit a function to exponential data using technology. Use the fitted function to solve problems

Additional 8th Grade Standards:

Work with radicals and integer exponents.	
NC.8.EE.1	Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

NC.8.EE.3	Use numbers expressed in scientific notation to estimate very large or very small quantities and to express how many times as much one is than the other.
NC.8.EE.4	Perform multiplication and division with numbers expressed in scientific notation to solve real-world problems, including problems where both decimal and scientific notation are used.

Unit 4: Introduction to Quadratic Functions & Equations	**≈ # of Weeks: 6**
Concepts to Integrate: Apply all to real world applications. • Interpret the structure of quadratic expressions. • Write quadratic expressions in equivalent forms to solve problems. • Perform the arithmetic operations on polynomials. • Understand the relationship between zeros and factors of quadratics. • Create quadratic equations that describe numbers or relationships. • Understand solving quadratic equations as a process of reasoning and explain the reasoning. • Solve quadratic equations in one variable. • Represent and solve quadratic equations graphically. • Understand the concept of a quadratic function and use function notation. • Interpret quadratic functions that arise in applications in terms of the context. • Analyze quadratic functions using different representations. • Construct and compare linear, exponential and quadratic models and solve problems.	
SMPs: *See Unit Planning Organizer*	

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

Understand the terms and properties of polynomials.	
NC.M1.A-APR.1	Perform arithmetic operations on polynomials. Build an understanding that operations with polynomials are comparable to operations with integers by adding and subtracting quadratic expressions and by adding, subtracting, and multiplying linear expressions.
NC.M1.A-SSE.1a	Interpret the structure of expressions. Interpret expressions that represent a quantity in terms of its context. Identify and interpret parts of a linear, exponential, or **quadratic** expression, including terms, factors, coefficients, and exponents.
NC.M1.A-SSE.1b	Interpret the structure of expressions. Interpret expressions that represent a quantity in terms of its context. Interpret a linear, exponential, or quadratic expression made of multiple parts as a combination of entities to give meaning to an expression.
Understand how changing the coefficients of a quadratic expression affect the key features of its related quadratic function.	
NC.M1.F-IF.7	Analyze functions using different representations. Analyze linear, exponential, and **quadratic** functions by generating different representations, by hand in simple cases and using technology for more complicated cases, to show key features, including: domain and range; rate of change; intercepts; intervals where the function is increasing, decreasing, positive, or negative; maximums and minimums; and end behavior.
NC.M1.F-IF.9	Analyze functions using different representations. Compare key features of two functions (linear, **quadratic**, or exponential) each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
NC.M1.F-LE.3	Construct and compare linear, exponential, and **quadratic** models and solve problems. Compare the end behavior of linear, exponential, and quadratic functions using graphs and tables to show that a quantity increasing exponentially exceeds a quantity increasing linearly or **quadratically**.
NC.M1.F-IF.6	Interpret functions that arise in applications in terms of the context. Calculate and interpret the average rate of change over a specified interval for a function presented numerically, graphically, and/or symbolically.
NC.M1.F-IF.8a	Analyze functions using different representations. Use equivalent expressions to reveal and explain different properties of a function. Rewrite a quadratic function to reveal and explain different key features of the function.
NC.M1.A-REI.10	Represent and solve equations and inequalities graphically. Understand that the graph of a two variable equation represents the set of all solutions to the equation.
NC.M1.F-IF2	Understand the concept of a function and use function notation. Use function notation to evaluate linear, **quadratic**, and exponential functions for inputs in their domains, and interpret statements that use function notation in terms of a context.
NC.M1.A-CED.2	Create and graph equations in two variables to represent linear exponential, and **quadratic** relationships between quantities.
NC.M1.A-CED.1	Create equations and inequalities in one variable that represent linear, exponential and **quadratic** relationships and use them to solve problems.
Understand the relationship between the factors of a quadratic expression and the solutions to its related quadratic equation.	
NC.M1.A-SSE.3	Write an equivalent form of a quadratic expression by factoring, where a is an integer of the quadratic expression, $ax^2 + bx + c$, to reveal the solutions of the equation or the zeros of the function the expression defines.
NC.M1.A-REI.4	Solve for the real solutions of quadratic equations in one variable by taking square roots and factoring.

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

NC.M1.A-APR.3	Understand the relationship between zeros and factors of polynomials. Understand the relationships among the factors of a quadratic expression, the solutions of a quadratic equation, and the zeros of a quadratic function.
NC.M1.A-REI.1	Understand solving equations as a process of reasoning and explain the reasoning. Justify a chosen solution method and each step of the solving process for linear and quadratic equations using mathematical reasoning.
NC.M1.A-REI.11	Build an understanding of why the x-coordinates of the points where the graphs of two linear, exponential, or **quadratic** equations $y = f(x)$ and $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$ and approximate solutions using a graphing technology or successive approximations with a table of values.
Build quadratic functions from other functions and interpret the key features of a quadratic function in context.	
NC.M1.F-IF.4	Interpret functions that arise in applications in terms of the context. Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities, including: intercepts; intervals where the function is increasing, decreasing, positive, or negative; and maximums and minimums.
NC.M1.A-APR.1	Perform arithmetic operations on polynomials. Build an understanding that operations with polynomials are comparable to operations with integers by adding and subtracting quadratic expressions and by adding, subtracting, and multiplying linear expressions.
NC.M1.A-SSE.1b	Interpret the structure of expressions. Interpret expressions that represent a quantity in terms of its context. Interpret a linear, exponential, or **quadratic** expression made of multiple parts as a combination of entities to give meaning to an expression.
NC.M1.F-BF.1b	Build a function that models a relationship between two quantities by combining linear, exponential, or **quadratic** functions with addition and subtraction or two linear functions with multiplication.
NC.M1.F-IF.5	Interpret a function in terms of context by relating its domain and range to its graph and, where applicable, to the quantitative relationship it describes
NC.M1.A-REI.11	Build an understanding of why the x-coordinates of the points where the graphs of two linear, exponential, or **quadratic** equations $y = f(x)$ and $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$ and approximate solutions using a graphing technology or successive approximations with a table of values.
NC.M1.F-IF.7	Analyze functions using different representations. Analyze linear, exponential, and **quadratic** functions by generating different representations, by hand in simple cases and using technology for more complicated cases, to show key features, including: domain and range; rate of change; intercepts; intervals where the function is increasing, decreasing, positive, or negative; maximums and minimums; and end behavior.

Unit 5: Systems of Equations & Inequalities	**≈ # of Weeks: 5**
NC Check-In #2	
Concepts to Integrate: Apply all to real world applications. • Create systems of linear equations that describe numbers or relationships.	

• Solve systems of linear equations. • Represent and solve systems of linear equations and inequalities graphically.
SMPs: *See Unit Planning Organizer*

Create, solve, and interpret systems of equations in context.	
NC.M1.A-CED.3	Create systems of **linear equations and inequalities** to model situations in context.
NC.M1.A-REI.5	Explain why replacing one equation in a system of linear equations by the sum of that equation and a multiple of the other produces a system with the same solutions.
NC.M1.A-REI.6	Use tables, graphs, or algebraic methods (substitution and elimination) to find approximate or exact solutions to systems of linear equations and interpret solutions in terms of a context.
NC.M1.A-REI.11	Build and understanding of why the x-coordinates of the points where the graphs of two linear, exponential, or quadratic equations $y = f(x)$ and $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$ and approximate solutions using a graphing technology or successive approximations with a table of values.
Create, solve, and interpret systems of inequalities in context.	
NC.M1.A-REI.12	Solve and represent the solutions of a linear inequality or a system of linear inequalities graphically as a region of the plane.

Additional 8th Grade Standards:

Analyze and solve linear equations and pairs of simultaneous linear equations.	
NC.8.EE.8	Analyze and solve a system of two linear equations in two variables in slope-intercept form. • Understand that solutions to a system of two linear equations correspond to the points of intersection of their graphs because the point of intersection satisfies both equations simultaneously. • Solve real-world and mathematical problems leading to systems of linear equations by graphing the equations. Solve simple cases by inspection.
NC.8.EE.7	Solve real-world and mathematical problems by writing and solving equations and inequalities in one variable. • Recognize linear equations in one variable as having one solution, infinitely many solutions, or no solutions. • Solve linear equations and inequalities including multi-step equations and inequalities with the same variable on both sides.

Unit 6: Descriptive Statistics	**≈ # of Weeks: 3**
Concepts to Integrate: Apply all to real world applications.	

NC Math 1 Scope and Sequence 2025-2026

Middle School

Refer to Unit Planning Organizers for Instructional Guidance

• Summarize, represent, and interpret data on a single count or measurement variable.
SMPs: *See Unit Planning Organizer*

Understand how to summarize, represent, interpret and compare data on a single count or measurement variable.	
NC.M1.S-ID.1	Use technology to represent data with plots on the real number line (histograms and box plots).
NC.M1.S-ID.2	Use statistics appropriate to the shape of the data distribution to compare center (median, mean) and spread (interquartile range, standard deviation) of two or more different data sets. Interpret differences in shape, center, and spread in the context of the data sets.
NC.M1.S-ID.3	Examine the effects of extreme data points (outliers) on shape, center, and/or spread.
Review/Extension/NC Math 1 EOC	

Additional 8th Grade Standards:

Work with radicals and integer exponents.	
NC.8.SP.4	Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table. • Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. • Use relative frequencies calculated for rows or columns to describe possible association between the two variables.